

Physics Horizons — Solution

X25 (2025) — English translation

A1

0.20 pts

Let the side of the square hole be $a = 1$ m. Find the characteristic angular dimension β of the hole as viewed from the screen. Compare it to the angular size of the Sun.

Let's calculate the angular size of the side of the square. Since $a \ll L$, then:

$$\beta \approx \frac{a}{L}.$$

Answer: \textbf{Answer:}

$$\beta \approx 6^\circ$$

A2

0.20 pts

Now let the side of the square hole be $a = 5$ mm. Find the characteristic angular dimension β of the hole as viewed from the screen. Compare it to the angular size of the Sun.

Similar to the previous point:

$$\beta \approx \frac{a}{L}.$$

Answer: \textbf{Answer:}

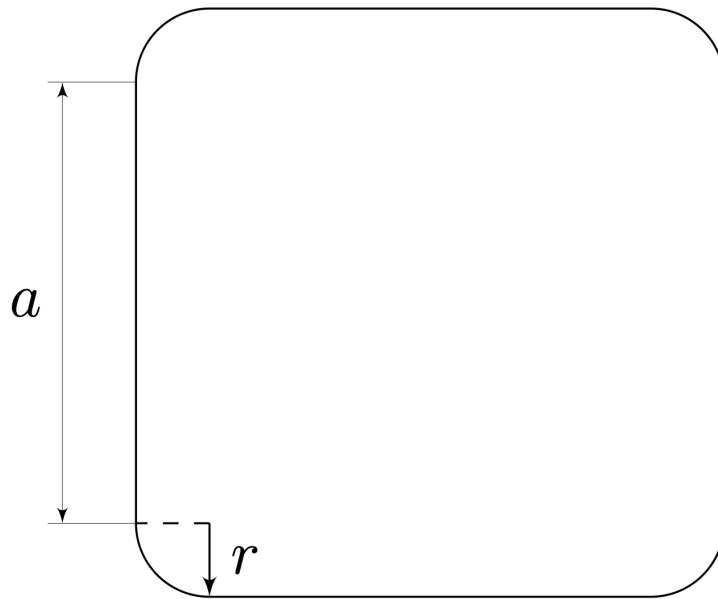
$$\beta \approx 0.03^\circ$$

A3

0.30 pts

In the answer sheets, qualitatively give the image of the Sun at $a = 1$ m. Please indicate the dimensions that characterize this image.

The image will look like a square with rounded corners. The straight part of the side has length a , the radius of curvature of the corners is $r = \frac{\alpha L}{2} = 4.4$ cm.

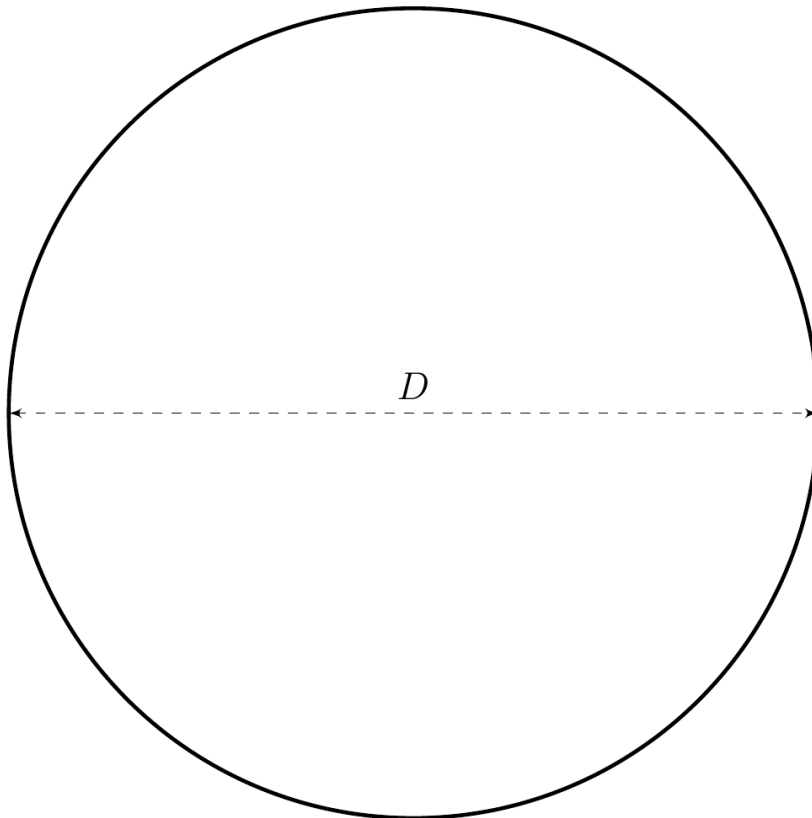


A4

0.50 pts

In the answer sheets, qualitatively give the image of the Sun at $a = 5$ mm. Please indicate the dimensions that characterize this image.

In this case, the image will differ slightly from a circle with diameter $D = \alpha L = 8.7$ cm.



B1

0.30 pts

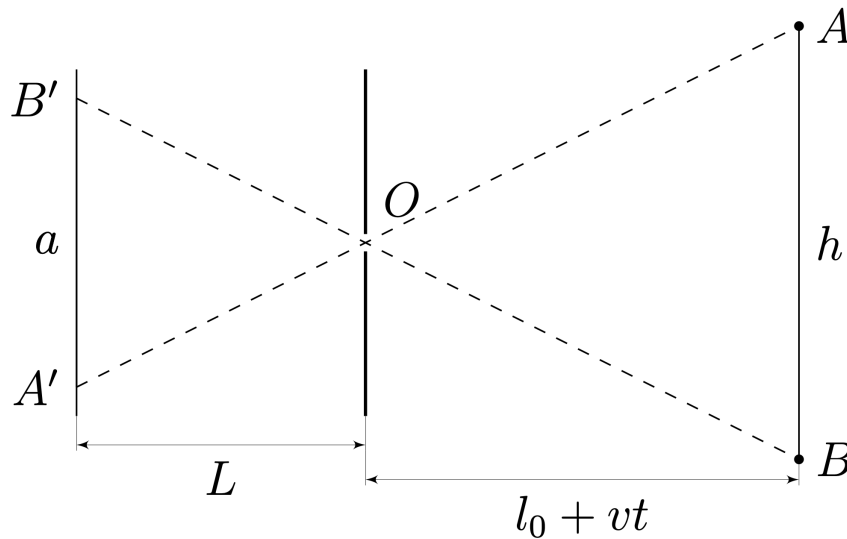
Find the dependence of the size a of a person's image on time t .

From the similarity of triangles AOB and $A'O'B'$:

$$\frac{a}{h} = \frac{L}{l_0 + vt}.$$

Answer: \textbf{Answer:}

$$a = \frac{Lh}{l_0 + vt}.$$



B2

0.70 pts

The figure shows images of an adult human and the Sun (angular size of the Sun $\alpha = 0.5^\circ$) at an unknown scale. Estimate how far away the adult was when the photograph was taken.

The angular sizes of the Sun and a person are of the same order, so we can assume that all the rays are paraxial and the sizes in the photograph are proportional to the angular sizes. From the figure, the ratio of the angular sizes of a person and the Sun:

$$\frac{h/l}{\alpha} = \frac{75\text{mm}}{30\text{mm}} = \frac{5}{2}.$$

For оценки возьмём $h = 2\text{m}$:

$$l = \frac{2h}{5\alpha} \approx 90 \text{ m}.$$

Answer: \textbf{Answer:}

$$l \approx 90 \text{ m}$$

C1

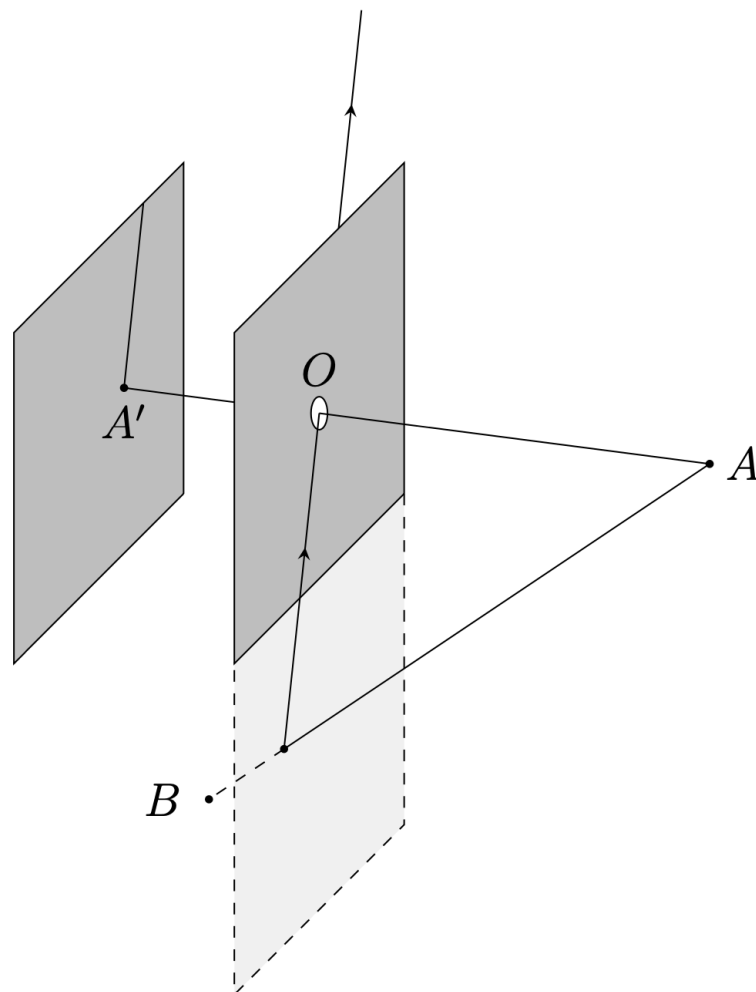
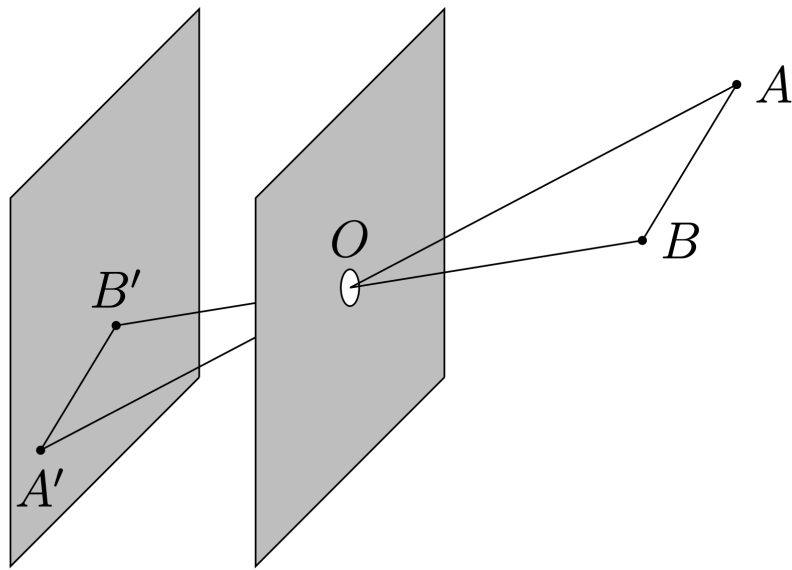
0.20 pts

Prove that the image of a segment in a photograph will be a line segment or a ray.

Note that A, B, O and the image of the segment lie in the same plane, the image AB lies entirely on the straight line - the intersection of this plane with the plane of the screen. If A and B are in front of the front plane of the camera, then the image will be a line segment.

If A and B are on different sides of the front plane of the camera, then the image will

be a ray.



At what ratio between l_0 and a will all the flag points be depicted on the screen at any time, if we neglect the finiteness of the screen dimensions? In all subsequent paragraphs of this part of the problem, consider the resulting condition fulfilled.

Some part of the flag will not be in the photo if it lies on the opposite side of the front plane of the camera. Let's consider what is happening in the projection onto a plane perpendicular to the plane of the flag and the front plane of the camera.

In order for the flag to be visible, it is always necessary that for any position of the point E (E is the center of the flag), the following must be fulfilled: $EC < EF$. The length of EC is constant:

$$EC = \frac{a\sqrt{2}}{2}.$$

EF is the hypotenuse of the right triangle EOF , therefore $EF = 2l_m$, where l_m is the length of the median of this triangle from the vertex O . The height of EOF from vertex O is l_0 , and it cannot be greater than l_m :

$$l_0 \leq l_m = \frac{EF}{2}.$$

That is $EF \geq 2l_0$ and equality выполняется, когда рассматриваемый треугольник равнобедренный. Thus, чтобы флаг всегда было видно, необходимо:

$$\frac{a}{\sqrt{2}} < 2l_0.$$

Answer: \textbf{Answer:}

$$a < 2\sqrt{2}l_0.$$

1.40 pts

The distance from the origin of coordinates XY to the image of a point is determined by the tangent of the angle between the camera axis and the direction to this point (for points A , B , C and D we denote these angles θ_A , θ_B , θ_C and θ_D , respectively). We can find the coordinates x , y and z of these points and get:

$$\tan \theta_A = \frac{|y_A|}{z_A} = \frac{\frac{a}{\sqrt{2}} \sin \xi}{\sqrt{l_0^2 + v^2 t^2} + \frac{a}{\sqrt{2}} \cos \xi};$$